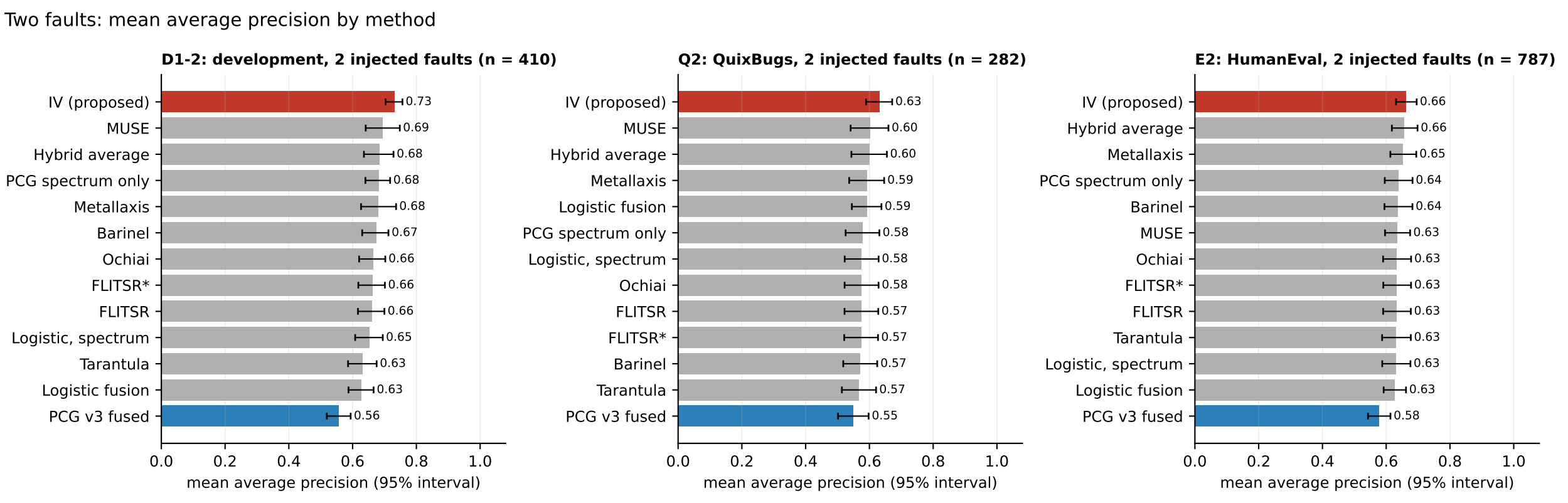
Two faults: mean average precision by method
D1-2: development, 2 injected faults (n = 410)
IV (proposed)
MUSE
Hybrid average
PCG spectrum only
Metallaxis
Barinel
Ochiai
FLITSR*
FLITSR
Logistic, spectrum
Tarantula
Logistic fusion
PCG v3 fused
0.73
0.69
0.68
0.68
0.68
0.67
0.66
0.66
0.66
0.65
0.63
0.63
0.56
0.0
0.2
0.4
0.6
0.8
1.0
mean average precision (95% interval)
Q2: QuixBugs, 2 injected faults (n = 282)
IV (proposed)
MUSE
Hybrid average
Metallaxis
Logistic fusion
PCG spectrum only
Logistic, spectrum
Ochiai
FLITSR
FLITSR*
Barinel
Tarantula
PCG v3 fused
0.63
0.60
0.60
0.59
0.59
0.58
0.58
0.58
0.57
0.57
0.57
0.57
0.55
0.0
0.2
0.4
0.6
0.8
1.0
mean average precision (95% interval)
E2: HumanEval, 2 injected faults (n = 787)
IV (proposed)
Hybrid average
Metallaxis
PCG spectrum only
Barinel
MUSE
Ochiai
FLITSR*
FLITSR
Tarantula
Logistic, spectrum
Logistic fusion
PCG v3 fused
0.66
0.66
0.65
0.64
0.64
0.63
0.63
0.63
0.63
0.63
0.63
0.63
0.58
0.0
0.2
0.4
0.6
0.8
1.0
mean average precision (95% interval)